C12.2 Synchronization message

The synchronization (Sync) message is used to ensure that all preceding DVM Operations are complete. On receipt of a Sync message, a component must ensure that:

- a TLB Invalidate operation is complete when a master can no longer use an invalidated translation and when all previous transactions that could have used an invalidated translation are complete
- a Branch Predictor Invalidate operation is complete when cached copies of predicted instruction fetches, from any virtual address or from a specified virtual address, have been invalidated and can no longer be accessed by the associated master
- an Instruction Cache Invalidate operation is complete when a master can no longer access a cached copy of the address locations that have been invalidated.

Note

The requirement that a TLB Invalidation is only complete when certain previous transaction are complete can, in some cases, require that the component that issued the transactions also issues a barrier transaction to be able to determine that the transactions have completed.

A component must have only one outstanding DVM Sync transaction. A component must receive a DVM Complete transaction before it issues another DVM Sync transaction.

Components must be able to accept DVM Sync messages and continue processing snoop transactions while waiting for earlier transactions to complete, that might be needed before a DVM Complete message can be sent. The maximum number of outstanding DVM Sync messages that a master must be able to accept is 256.

A DVM Sync must complete in a timely manner, even if the component continues to receive more DVM Operations and more DVM Sync messages.

It is not acceptable for a component, which has received a DVM Sync message, to continue to issue transactions if further DVM Operations after the DVM Sync mean that the newly issued transactions must complete before the DVM Sync can complete.

Note

Support for multiple outstanding DVM Sync messages only requires the component to be aware of the number of DVM Complete responses required. No additional information about the individual DVM Sync messages is necessary.

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